21-242 Recitation 12

RREF — Q5. Same image without A already in RREF

Course-level solution, with a later-course alternative when useful

October 1, 2026

Prompt

Is it possible that $\operatorname{im}(T_A) = \operatorname{im}(T_{\mathrm{RREF}(A)})$ but $A \neq \mathrm{RREF}(A)$?

Course-level solution

Yes. Take

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}, \qquad \mathrm{RREF}(A) = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

The column spaces are both $\mathrm{Span}\{(1,0)^T\}$, but $A \neq \mathrm{RREF}(A)$.
(The phenomenon is: scaling a pivot column of A does not change its span, but does change the matrix.)